

FIN305 Risk Management for Business

Week 3-4 Questions

Practice questions and problems

The numbers in parentheses after the questions correspond to the original numbering used in the textbook, which is intended to assist students in easily finding the relevant questions within the book.

1. What is the difference between expected shortfall and VaR? What is the theoretical advantage of expected shortfall over VaR? (11.1)
2. A fund manager announces that the fund's one-month 95% VaR is 6% of the size of the portfolio being managed. You have an investment of \$100,000 in the fund. How do you interpret the portfolio manager's announcement? (11.3)
3. Suppose that each of two investments has a 0.9% chance of a loss of \$10 million and a 99.1% chance of a loss of \$1 million. The investments are independent of each other. (11.5)
 - (a) What is the VaR for one of the investments when the confidence level is 99%?
 - (b) What is the expected shortfall for one of the investments when the confidence level is 99%?
 - (c) What is the VaR for a portfolio consisting of the two investments when the confidence level is 99%?
 - (d) What is the expected shortfall for a portfolio consisting of the two investments when the confidence level is 99%?

- (e) Show that in this example VaR does not satisfy the subadditivity condition, whereas expected shortfall does.
4. Suppose that we back-test a VaR model using 1,000 days of data. The VaR confidence level is 99% and we observe 17 exceptions. Should we reject the model at the 5% confidence level? Use a one-tailed test. (11.9)
 5. The change in the value of a portfolio in three months is normally distributed with a mean of \$500,000 and a standard deviation of \$3 million. Calculate the VaR for a confidence level of 99.5% and a time horizon of three months. (11.16)
 6. The probability that the loss from a portfolio will be greater than \$10 million in one month is estimated to be 5%. What is the one-month 99% VaR assuming the change in value of the portfolio is normally distributed with zero mean? (11.17)
 7. What assumption is being made when VaR is calculated using the historical simulation approach and 500 days of data? (12.1)

Solutions

Each week, students should attempt the provided exercise questions after lectures. Solutions can be found at the end of the textbook to check your answers. For those questions to which no solutions can be found in the textbook, the detailed solutions/explanations will be shown in the present file as follows. Selected questions will be discussed in tutorials, where the instructor will offer more detailed solutions. Recordings of the related tutorials will be made available to all students via Learning Mall.

Question (5)

- The loss has a mean of -500 and a standard deviation of 3,000. Also, $N^{-1}(0.995) = 2.576$. The 99.5% VaR in \$000s is $-500 + 3,000 \times 2.576 = 7227$. We are 99.5% certain that the loss will not be greater than \$7.227 million.

Question (6)

- The 5% tail of the distribution starts 1.645 standard deviations from the mean. The standard deviation is therefore $10/1.645 = \$6.080$ million. The one-month 99% VaR is therefore $6.080 \times N - 1(0.99) = \14.143 million.